

Automated Traffic Violation Detection and Identification System Using Computer Vision

Team Members

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Executive Summary

Traffic surveillance systems generate a massive volume of photographic evidence every day. Manual inspection of these images for identifying traffic violations is time-consuming, resource-intensive, and prone to inconsistencies.

This proposal presents an AI-powered Traffic Violation Detection and Identification System that automatically analyzes traffic images, detects vehicles and road users, identifies traffic violations, extracts vehicle registration details using OCR, generates annotated evidence, and produces actionable reports for traffic authorities.

To validate the feasibility of the proposed solution, a proof-of-concept prototype has already been developed and tested. The current prototype focuses on motorcycle-related violations and demonstrates promising performance in automated violation detection and license plate recognition.

The proposed architecture is scalable and can be extended to support additional vehicle categories and violation types required for large-scale deployment.

Problem Statement

Modern cities deploy large networks of traffic surveillance cameras to improve road safety and traffic management. However, the manual review of captured images remains a significant bottleneck.

Challenges include:

- Large volume of traffic images.
- Human-intensive violation inspection.
- Delayed enforcement actions.
- Inconsistent identification of violations.
- Difficulty in generating searchable records and analytics.

An intelligent computer vision system can automate these tasks and significantly improve traffic monitoring efficiency.

Proposed Solution

We propose a scalable Computer Vision and OCR based framework capable of automatically processing traffic images and detecting traffic violations.

The system performs:

- Image enhancement and preprocessing.
- Vehicle and road user detection.
- Vehicle classification.
- Traffic violation identification.
- License plate recognition.
- Evidence generation.
- Analytics and reporting.

The framework is designed to operate under varying traffic densities, lighting conditions, and environmental challenges.

System Architecture

Methodology

1. Image Preprocessing

The system first enhances incoming traffic images to improve detection performance.

Techniques include:

- Contrast enhancement
 - Noise reduction
 - Image normalization
 - Blur mitigation
 - Shadow handling
 - Low-light enhancement
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2. Vehicle and Road User Detection

Deep learning models detect:

- Motorcycles
- Cars
- Buses
- Trucks
- Riders
- Drivers
- Pedestrians

Detected objects are localized using bounding boxes.

3. Traffic Violation Detection

The violation engine analyzes detected vehicles and road users to identify violations.

Implemented in Prototype

- Helmet Non-Compliance Detection
- Triple Riding Detection

Proposed Extensions

- Seatbelt Non-Compliance Detection
- Wrong-Side Driving Detection
- Stop-Line Violation Detection
- Red-Light Violation Detection
- Illegal Parking Detection

Each violation is assigned a confidence score to support decision-making.

4. License Plate Recognition

The system performs:

- Number plate localization
- Plate image extraction
- Optical Character Recognition (OCR)
- Registration number extraction

This enables automated vehicle identification and record generation.

5. Evidence Generation

For every detected violation, the system generates:

- Annotated violation image
- Vehicle details
- Extracted registration number
- Timestamp
- Violation category

The generated evidence can be reviewed by traffic authorities.

6. Analytics and Reporting

The system maintains searchable violation records and supports:

- Violation statistics
- Trend analysis
- Vehicle-wise search

- Evidence retrieval
 - Summary reporting
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Prototype Implementation

To validate the proposed framework, a proof-of-concept prototype has been developed.

Implemented Components

Vehicle Detection and Classification

Detects:

- Motorcycle
- Car
- Bus
- Truck

Helmet Violation Detection

Identifies riders without helmets.

Triple Riding Detection

Detects motorcycles carrying three or more riders.

License Plate Detection

Detects vehicle number plates from traffic images.

OCR-Based Number Plate Recognition

Extracts registration details using EasyOCR.

Evidence Generation

Produces annotated evidence images.

Dashboard Visualization

Displays violation records and evidence images.

Technology Stack

Component	Technology
Deep Learning	YOLOv8
Computer Vision	OpenCV
OCR	EasyOCR
Data Processing	Pandas, NumPy
Dashboard	Flask
Development Environment	Python, JupyterLab

Experimental Results

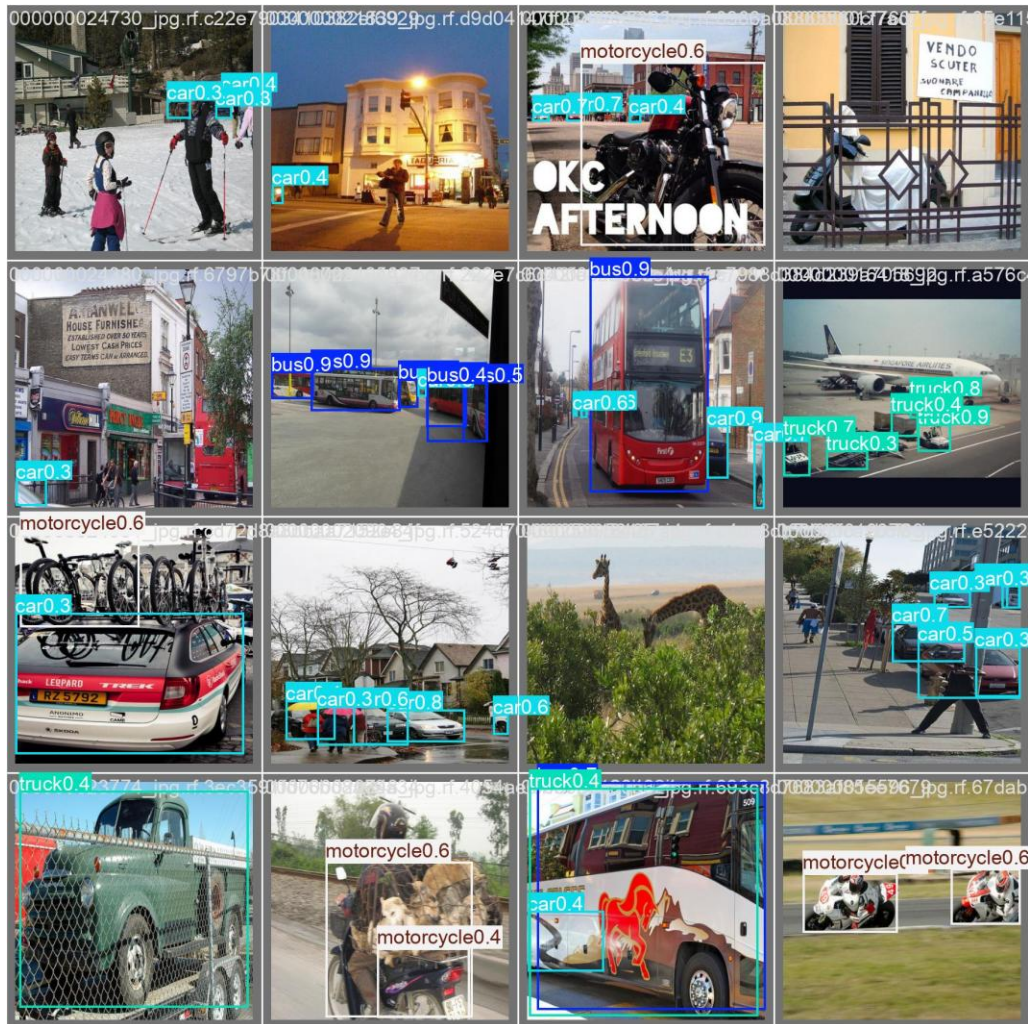
The proof-of-concept prototype was evaluated using multiple custom-trained YOLOv8 models.

Model	Precision	Recall	mAP50	mAP50-95
Vehicle Detection	0.766	0.795	0.622	0.370
Helmet Detection	0.713	0.778	0.788	0.445
Triple Riding Detection	0.929	0.876	0.953	0.873
License Plate Detection	0.950	0.844	0.903	0.639

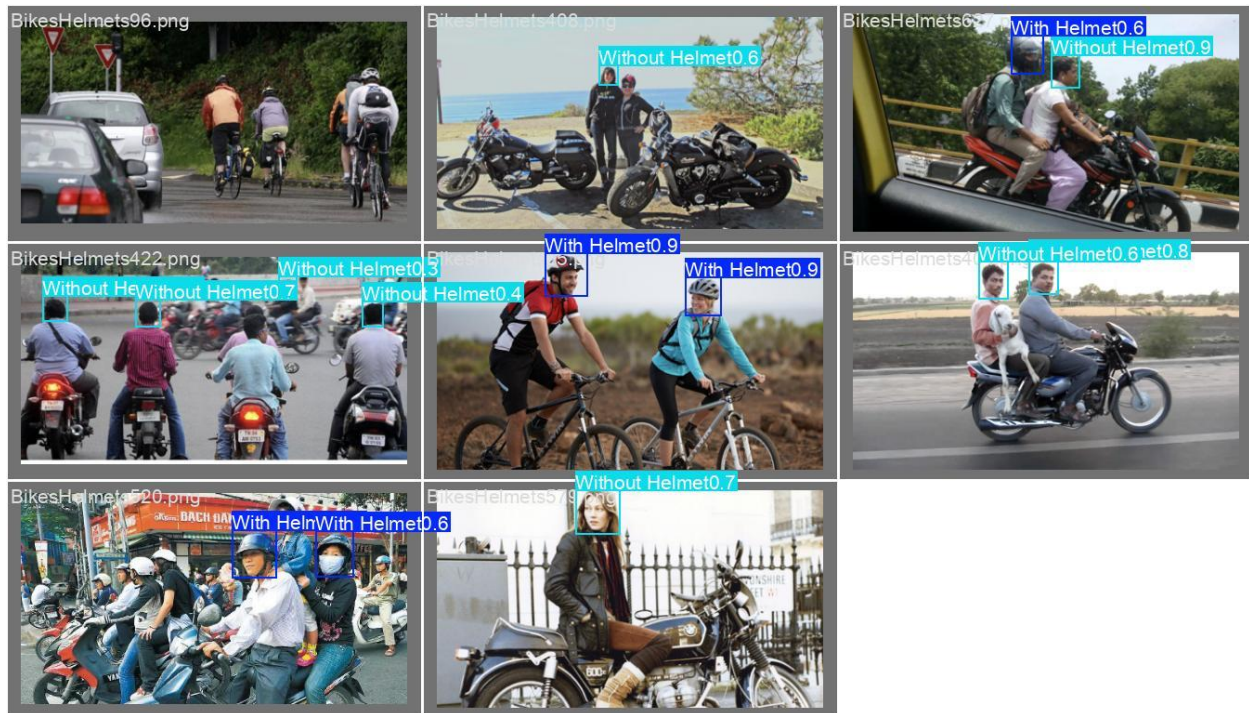
These results demonstrate the feasibility of the proposed approach and indicate strong performance for motorcycle-focused traffic violation detection.

Prototype Validation

Vehicle Detection Output



Helmet Violation Detection Output



Triple Riding Detection Output



License Plate Recognition Output



Dashboard Output

Traffic Violation Dashboard						
6 Total Records		6 Helmet Violations		3 Triple Riding Violations		4 Unique Vehicles
Timestamp	Vehicle Type	Plate Number	Violation Type	Helmet	Triple	Evidence
2026-06-21 13:09:16.698915	motorcycle	UNKNOWN	Helmet	True	False	View Evidence
2026-06-21 13:09:16.452809	motorcycle	UNKNOWN	Helmet + Triple Riding	True	True	View Evidence
2026-06-21 13:09:16.166972	motorcycle	UNKNOWN	Helmet + Triple Riding	True	True	View Evidence
2026-06-21 13:09:15.744080	motorcycle	UP16Y545	Helmet + Triple Riding	True	True	View Evidence
2026-06-21 13:09:15.325461	motorcycle	LER9997	Helmet	True	False	View Evidence
2026-06-21 13:09:15.140820	motorcycle	LER997	Helmet	True	False	View Evidence



Helmet status: Without helmet

Triple status: Triple riding

Scalability and Future Scope

The modular architecture allows seamless expansion beyond motorcycle-based violations.

Future enhancements include:

- Extension to all vehicle categories including cars, buses, trucks, and commercial vehicles.
- Real-time CCTV video processing.
- Seatbelt violation detection.

- Wrong-side driving detection.
 - Stop-line violation detection.
 - Red-light violation detection.
 - Illegal parking detection.
 - Multi-camera traffic monitoring.
 - Automated challan generation.
 - Cloud deployment.
 - Traffic analytics dashboard.
 - Smart city integration.
 - Citizen grievance and appeal portal for disputed violations.
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Expected Impact

The proposed system can significantly improve traffic law enforcement by:

- Reducing manual monitoring effort.
 - Increasing violation detection consistency.
 - Improving enforcement efficiency.
 - Generating reliable digital evidence.
 - Supporting large-scale smart-city deployments.
 - Enhancing road safety through automated monitoring.
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Conclusion

This proposal presents a scalable AI-driven framework for automated traffic violation detection and classification using Computer Vision and OCR technologies.

A working proof-of-concept prototype has already been implemented and evaluated, demonstrating promising results for motorcycle-focused traffic violation detection, license plate recognition, evidence generation, and dashboard visualization.

The architecture is designed for future expansion to support additional vehicle categories and violation types, making it a practical and scalable solution for intelligent traffic monitoring and enforcement.